@NEETXPR

17/01/2024

Code-A_Phase 1

Aakash + BYJU'S

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MM : 720 NCERT Booster Test Series-2023-24_Test-04A **Time : 200 Min.**

Topics Covered:
Physics : Laws of Motion, Work, Energy & Power, System of Particles & Rotational Motion
Chemistry : Chemical Bonding and Molecular Structure, The p-Block Elements (Group-15 to 16), The d and f-Block Elements
Botany : The Living World, Sexual Reproduction in Flowering Plants, Principles of Inheritance and Variation, Biological Classification
Zoology : Biomolecules-II, Evolution, Breathing and Exchange of Gases

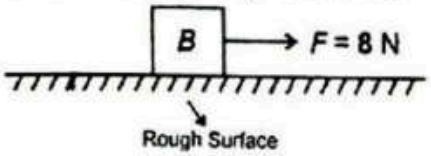
General Instructions :

1. There are two sections in each subject, i.e. **Section-A & Section-B.** You have to attempt **all 35 questions** from Section-A & only **10 questions** from Section-B out of **15.**
2. Each question carries **+4 marks.** For every wrong response, **–1 mark** shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
3. Use blue/black ballpoint pen only to darken the appropriate circle.
4. Mark should be dark and completely fill the circle.
5. Dark only one circle for each entry.
6. Dark the circle in the space provided only.
7. Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

SECTION-A

1. In the figure shown if limiting friction for block is 10 N then friction acting on the block is



(1) 10 N
(2) 5 N
(3) 8 N
(4) 2 N

2. If a force is applied to a body externally, then it's momentum's
(1) Magnitude must vary
(2) Direction must vary
(3) Magnitude may remain constant
(4) Magnitude must remain constant

3. The dimension of linear impulse is given as
(1) $[M^0L^{-1}T^{-2}]$
(2) $[LT^{-1}]$
(3) $[ML^1T^{-1}]$
(4) $[M^{-1}L^2T^{-3}]$

4. The relationship $\frac{dp}{dt} = m\frac{dv}{dt}$ where symbols have usual meaning is valid for
(1) System of variable mass
(2) System of fixed mass
(3) Any kind of system
(4) No system

5. The potential energy of a system decreases if
(1) Conservative forces do positive work on the system
(2) Conservative forces do negative work on the system
(3) Conservative forces do no work on the system
(4) Both (2) and (3)

6. A body at rest can have
(1) Kinetic energy
(2) Momentum
(3) Potential energy
(4) Both (1) & (2)

7. If a vector $\hat{i} + \hat{j} + \hat{k}$ is perpendicular to the vector $a\hat{i} + \hat{j} - \hat{k}$, then value of the constant a is
(1) Zero
(2) 1
(3) 2
(4) 3

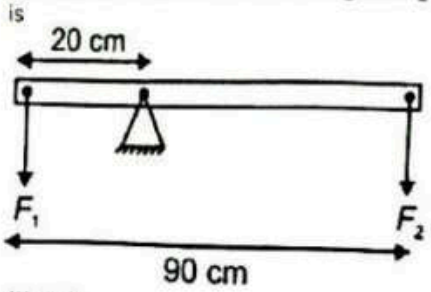
8. A particle moves along x axis from $x = 0$ to $x = 2$ m under the influence of force $F = (7 - 2x + 3x^2)$N. The work done during this is
(1) 18 J
(2) 30 J
(3) 25 J
(4) 10 J

9. A ball is dropped from height 10 m on the ground. If coefficient of restitution is 0.5, then the height of the ball after 2nd bounce is
(1) 0.625 m
(2) 1.25 m
(3) 2.5 m
(4) 5 m

10. **Statement A:** Energy is cause for doing work and work is the effect.
Statement B: Energy is scalar quantity but power is a vector quantity.
(1) A and B are correct
(2) A and B are incorrect
(3) A is correct and B is incorrect
(4) A is incorrect and B is correct

11. The angular velocity of a body is $\vec{\omega} = \left(2\hat{i} + \hat{j} + 5\hat{k}\right)$ rad/s under a torque $\vec{\tau} = \left(5\hat{i} + 2\hat{j} + 3\hat{k}\right)$ N-m acting on it. The instantaneous rotational power is
(1) 27 W
(2) 10 W
(3) 18 W
(4) 24 W

12. A light rod of length 90 cm is placed as shown. If two forces F_1 and F_2 is applied for equilibrium as shown, the ratio of F_1 and F_2 is



(1) 2 : 9
(2) 9 : 2
(3) 4 : 7
(4) 7 : 2

13. The moment of inertia of a body depends on
(1) The mass of body
(2) The distribution of mass in the body
(3) The position of axis of rotation of body
(4) All of these

14. Force couple acting on a rigid body produces
(1) Linear motion
(2) Rotational motion
(3) Vibrational motion
(4) Both (1) and (2)

15. A particle of mass 2 kg located at the position $\left(\hat{i} + \hat{j}\right)$m has a velocity $2\left(\hat{i} - \hat{j} + \hat{k}\right)$m/s. Z-component of its angular momentum about origin is
(1) Zero
(2) +8
(3) 12
(4) −8

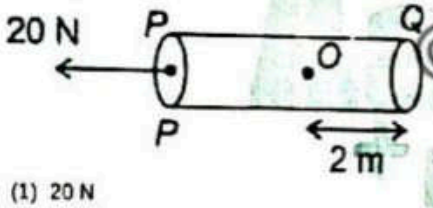
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16. A force $\vec{F} = \left(2\hat{i} + 4\hat{j} + 6\hat{k}\right)$ N displaces a particle from P(2, 3, 4) to Q(5, 6, 8), then the work done by this force is
(1) 20 J
(2) 42 J
(3) 40 J
(4) 24 J

17. If angle between the force acting on a body and displacement of the body is obtuse then the work done by the force is
(1) Positive
(2) Negative
(3) Infinite
(4) Both (1) and (3) are possible

18. The centre of mass of two masses m_1 and m_2 moves by a distance $\frac{x}{6}$, when mass m_1 is moved by distance x and m_2 is kept fixed. The ratio m_2/m_1 is
(1) 4 : 1
(2) 3 : 1
(3) 6 : 1
(4) 5 : 1

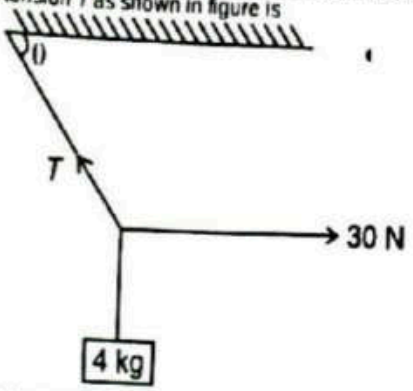
19. Consider a uniform rope PQ of length 20 m as shown in figure. The value of tension force acts at point O if 20 N force acts at end P of rope



(1) 20 N
(2) 2 N
(3) 15 N
(4) 10 N

20. If the external force acting on a body has zero resultant, the body
(1) Must not move
(2) Must move
(3) Must not accelerate
(4) May accelerate

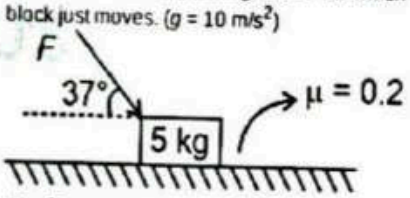
21. If the given system is in equilibrium then tension T as shown in figure is



(1) 50 N
(2) 30 N
(3) 40 N
(4) 65 N

22. Choose the correct statement with regard to the limiting friction.
(1) Direction of force of limiting friction is always opposite to the direction in which the body is on the verge of moving over the other
(2) It depends on nature of material of the surfaces in contact
(3) It is independent of apparent contact area, as long as normal reaction between two bodies in contact remains same
(4) All of these

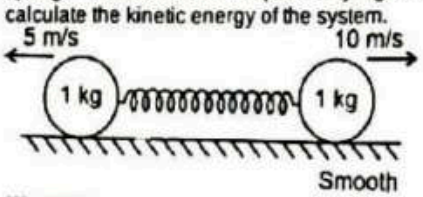
23. A block of mass 5 kg is placed on a rough table of coefficient of friction 0.2 as shown in figure. Find out the pushing force F for which block just moves. (g = 10 m/s^2)



(1) $\frac{25}{7}$ N
(2) $\frac{30}{7}$ N
(3) $\frac{50}{7}$ N
(4) $\frac{250}{17}$ N

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24. Two particles are connected with a light spring. For the instant depicted by figure calculate the kinetic energy of the system.

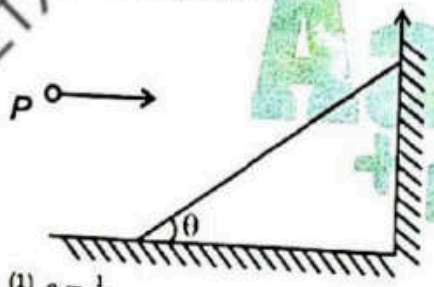


(1) 22.5 J
(2) 62.5 J
(3) 20 J
(4) 44.5 J

25. A : The work done by spring force in closed path is always zero.
R : Spring force is conservative in nature.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

26. A particle P strikes the inclined smooth plane horizontally and rebounds vertically as shown in figure. If the angle θ is 60° then coefficient of restitution is



(1) $e = \frac{1}{2}$
(2) $e = \frac{1}{3}$
(3) $e = \frac{1}{4}$
(4) $e = \frac{1}{5}$

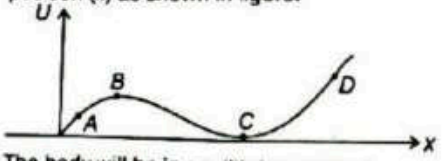
27. Potential energy of a particle is given as $U = 2x^2 - 5$. The value of force at $x = 2$ m is

(1) 8 N
(2) 4 N
(3) 5 N
(4) 6 N

28. A unit vector perpendicular to each of the vectors $2\hat{i} + 4\hat{j} - 6\hat{k}$ and $2\hat{i} - 4\hat{j} + 2\hat{k}$ is

(1) $2\left(\hat{i} + \hat{j} + 2\hat{k}\right)$
(2) $\sqrt{3}\left(\hat{i} + \hat{j} + \hat{k}\right)$
(3) $\frac{\sqrt{3}}{2}\left(\hat{i} + \hat{j} - \hat{k}\right)$
(4) $-\frac{1}{\sqrt{3}}\left(\hat{i} + \hat{j} + \hat{k}\right)$

29. The variation of potential energy U, of a body moving along x-axis, varies with its position (x) as shown in figure.



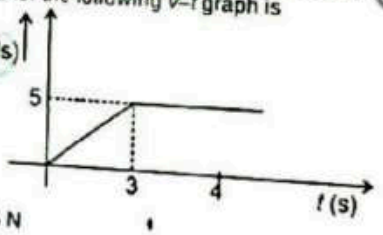
The body will be in equilibrium state at

(1) A and B
(2) B and C
(3) C and A
(4) D and B

30. Choose the option representing the correct relation between units.

(1) 1 erg = 10^7 joule
(2) 1 joule = 10^7 erg
(3) 1 joule = 10^5 erg
(4) 1 erg = 10^5 joule

31. The force acting on a particle of mass 5 kg at $t = 4$ s for the following v–t graph is



(1) 5 N
(2) Zero
(3) 20 N
(4) 15 N

32. A conveyer belt is moving with constant velocity 3 m/s. If sand is poured at the rate of 9 kg/s on the belt then the force required to move the belt with same constant velocity is

(1) 10 N
(2) 12 N
(3) 27 N
(4) Zero

33. The momentum of a body depends on time as $P = (2t^3 + t^2 - 1)$ Ns. The force acting on the body at $t = 2$ s is
(1) 20 N
(2) 28 N
(3) 2 N
(4) 14 N

34. An engine pumps water of density ρ through a hose pipe. Water leaves the hose pipe with a velocity v. Find the rate at which kinetic energy is imparted to water
(1) $\frac{1}{2}\rho A v^3$
(2) $\frac{1}{2}\rho A v$
(3) $\rho A v$
(4) $\frac{1}{3}\rho A v^2$

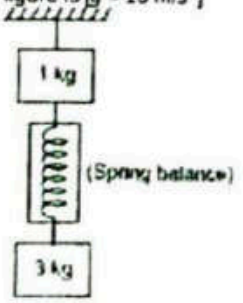
35. A : In an inelastic collision between two bodies, the relative speed of the bodies after collision is less than the relative speed before the collision.
R : In an inelastic collision linear momentum of the system remain conserved.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

SECTION-B

36. According to Newton's 3rd law of motion
(1) Action arises before reaction
(2) Reaction arises before action
(3) Action and reaction always act on same bodies
(4) Action and reaction always act on different bodies

38. The value of coefficient of restitution for perfectly elastic collision is
(1) 1
(2) Zero
(3) $\frac{1}{2}$
(4) $\frac{1}{\sqrt{2}}$

37. The reading of spring balance as shown in figure is [$g = 10$ m/s^2]



(1) 30 N
(2) 20 N
(3) 40 N
(4) 10 N

39. A body with mass 1 kg moves with velocity $\vec{v} = \left(2\hat{i} + 4\hat{j}\right)$ ms^{-1}. Then its kinetic energy is
(1) 8 J
(2) 12 J
(3) 10 J
(4) 6 J

40. If a body is released from a certain height, during its fall
(1) Both potential and kinetic energy of the body decrease
(2) Both potential and kinetic energy of the body increase
(3) Its potential energy increases and kinetic energy decreases
(4) Its kinetic energy increases and potential energy decreases

41. The work done by the gravity on a body moving upwards is
(1) Zero
(2) Negative
(3) Positive
(4) Can be zero or positive

42. Moment of inertia of a uniform cylinder of mass M and radius R about its geometric axis is
(1) $\frac{1}{2}MR^2$
(2) $\frac{1}{4}MR^2$
(3) MR^2
(4) $\frac{3}{2}MR^2$

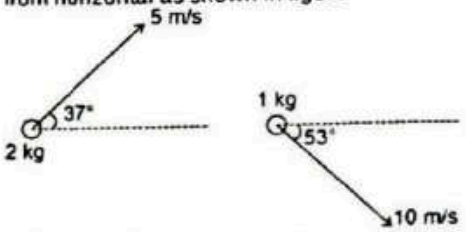
43. A disc of radius 2 m and mass 100 kg rolls on a horizontal floor. Its centre of mass has speed of 20 cm/s. How much work is needed to stop it?
(1) 3 J
(2) 30 kJ
(3) 2 J
(4) 1 J

44. The moment of inertia of a flywheel is 4 kg m^2. What angular acceleration will be produced in it by applying a torque of 10 N-m on it?
(1) 2.5 rad/s^2
(2) 5 ra 1/s^2
(3) 3 rad/s^2
(4) 2 rad/s^2

45. The torque that must be applied to a 50 kg flywheel of radius of gyration 40 cm to give it an angular acceleration of 2 rad/s^2 is
(1) 8 Nm
(2) 16 Nm
(3) 6 Nm
(4) 18 Nm

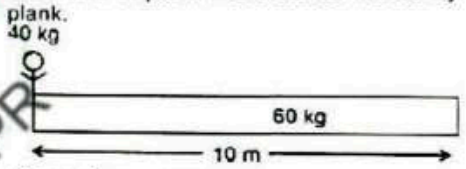
46. The minimum velocity with which a body of mass m must enter a vertical circular loop of radius R at the lowest point of loop so that it can complete the loop, is
(1) $\sqrt{8gR}$
(2) $\sqrt{gR}$
(3) $\sqrt{5gR}$
(4) $\sqrt{3gR}$

47. A particle of mass 2 kg moves with velocity 5 m/s in a direction 37° from horizontal and 1 kg with velocity 10 m/s in a direction 53° from horizontal as shown in figure



Calculate velocity of centre of mass.
(1) $\left(\frac{14}{3}\hat{i} - \frac{2}{3}\hat{j}\right)$ m/s
(2) $\left(\frac{2}{3}\hat{i} - \frac{14}{3}\hat{j}\right)$ m/s
(3) $\left(\frac{5}{3}\hat{i} - \frac{7}{3}\hat{j}\right)$ m/s
(4) $\left(\frac{7}{3}\hat{i} - \frac{5}{3}\hat{j}\right)$ m/s

48. A person of mass 40 kg is standing on a plank of length 10 m. If person moves on the other side of plank find distance moved by plank.



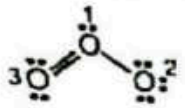
(1) $x = 2$ m
(2) $x = 4$ m
(3) $x = 5$ m
(4) $x = 1$ m

49. Work done by friction is
(1) Always zero
(2) Always positive
(3) May be positive, negative or zero
(4) Always negative

50. The coefficient of friction between a body and ground is 0.75 then, the angle of friction can vary from
(1) 0° to 90°
(2) 0° to 37°
(3) 0° to 53°
(4) 37° to 53°

CHEMISTRY

51. Formal charge on oxygen atoms 1 and 2 respectively in the following structure is



(1) 1 and zero
(2) 1 and 2
(3) –1 and 1
(4) 1 and –1

52. Which of the following molecules has highest dipole moment?
(1) NF_3
(2) NH_3
(3) BF_3
(4) CH_4

53. The ratio of σ bonds and π bonds in Benzene is
(1) 2 : 1
(2) 4 : 1
(3) 4 : 3
(4) 5 : 3

54. Incorrect statement is
(1) H_2O_2 molecule contains both polar and non-polar bonds
(2) XeO_4 contains equal number of σ bonds and π bonds, all the 'π' bonds are $p\pi - d\pi$ bonds
(3) Number of 109°28' angles in CH_4 are six
(4) In PCl_5 molecules all 'P-Cl' bond length are equal

55. Which among the following molecule is not planar?
(1) $BeCl_2$
(2) SO_3
(3) XeF_4
(4) SF_4

56. Which of the following molecules has an odd electron?
(1) H_2O
(2) CH_4
(3) NH_3
(4) NO

SECTION-A

57. In the hydrocarbon, $\overset{4}{C}H_3 - \overset{3}{C}H = \overset{2}{C} = \overset{1}{C}H_2$, the state of hybridisation of carbon atoms 1, 2 and 4 respectively are
(1) sp^2, sp, sp^3
(2) sp^2, sp^2, sp^3
(3) sp, sp, sp^3
(4) sp^2, sp, sp^2

58. The hybridisation of PCl_5 in solid state
(1) sp^3d
(2) sp^3d^2
(3) sp^3 and sp^3d^2
(4) sp^3 and sp^2

59. In given molecules highest % s-character of C atom is observed in
(1) C_2H_2
(2) C_2H_4
(3) C_3H_6
(4) C_3H_8

60. The hybridisation state of central atom in SF_6 is
(1) sp^3d^2
(2) sp^3d^3
(3) dsp^2
(4) sp^3

61. Hydride of group 15 elements of highest boiling point is
(1) NH_3
(2) PH_3
(3) SbH_3
(4) BiH_3

62. Hydride of group-15, of highest reducing character is
(1) NH_3
(2) BiH_3
(3) SbH_3
(4) PH_3

63. On heating $(NH_4)_2Cr_2O_7$, which of the following is not formed?
(1) NH_3
(2) N_2
(3) H_2O
(4) Cr_2O_3

64. Catalyst used in Haber process for the synthesis of ammonia is
(1) Co
(2) Fe
(3) Ni
(4) Cu

65. The neutral oxide of Nitrogen is
(1) N_2O_5
(2) N_2O
(3) N_2O_4
(4) NO_2

66. Which among the following is not formed during Ostwald's process?
(1) N_2O
(2) NO
(3) NO_2
(4) HNO_3

67. The P–P–P bond angle in white phosphorous is
(1) 90°
(2) 180°
(3) 60°
(4) 120°

68. Which of the following is not a transition element?
(1) Sc
(2) Zn
(3) Cu
(4) Mn

69. Number of unpaired e^- in Cr^{2+} ion is
(1) 1
(2) 2
(3) 3
(4) 4

70. Match List-I with List-II

List-I Ions		List-II Magnetic moment in BM	
(A)	Mn^{2+}	(P)	$\sqrt{24}$
(B)	Zn^{2+}	(Q)	$\sqrt{35}$
(C)	Cu^{2+}	(R)	$\sqrt{3}$
(D)	Fe^{2+}	(S)	0

Choose the correct option
(1) (A) → P; (B) → Q; (C) → R; (D) → S
(2) (A) → Q; (B) → S; (C) → R; (D) → P
(3) (A) → S; (B) → R; (C) → Q; (D) → P
(4) (A) → Q; (B) → P; (C) → S; (D) → R

71. Enthalpy of atomization is minimum for
(1) Ti
(2) Cr
(3) Mn
(4) Co

72. Which oxide of Mn is most acidic in nature?
(1) MnO
(2) MnO_2
(3) Mn_2O_7
(4) Mn_2O_3

73. The acidic strength of oxyacids follow the order
(1) $HClO_3 > HClO_2 > HOCl$
(2) $HClO_3 < HClO_2 < HOCl$
(3) $HClO_3 > HClO_2 < HOCl$
(4) $HClO_3 < HClO_2 > HOCl$

74. Which of the following statements regarding conc. sulphuric acid is not correct?
(1) It is reduced to SO_2 on reaction with Cu.
(2) It is a strong dehydrating agent
(3) It can be used to manufacture more volatile acids from their corresponding salts
(4) The absorption of SO_2 in sulphuric acid produces oleum

75. Iodide ions oxidise by ozone into
(1) IO^-
(2) IO_3^-
(3) I_2
(4) IO_2^-

76. When hot and concentrated NaOH reacts with Cl_2, the major products formed are
(1) NaCl + NaClO
(2) NaCl + $NaClO_3$
(3) NaClO + $NaClO_3$
(4) NaCl + $NaClO_2$

77. In which of the following compounds, peroxide linkage is present?
(1) $H_2S_4O_6$
(2) H_2SO_3
(3) $H_2S_2O_8$
(4) $H_2S_2O_7$

78. Which of the following has bleaching action due to its reducing property?
(1) O_3
(2) H_2O_2
(3) Cl_2
(4) SO_2

79. The species which is green in colour is
(1) MnO_4^-
(2) CrO_4^{2-}
(3) MnO_4^{2-}
(4) $Cr_2O_7^{2-}$

80. Most basic hydroxides among the following is
(1) $Eu(OH)_3$
(2) $Er(OH)_3$
(3) $Tm(OH)_3$
(4) $Yb(OH)_3$

81. Actinoid contraction is more than lanthanoid contraction because
(1) Energy of 4*f* and 5*f* orbitals are same
(2) 5*f* orbitals are more diffused as compared to 4*f* orbitals
(3) Actinoids show greater range of oxidation states than lanthanoids
(4) Actinoids are highly reactive

82. Potassium manganate (K_2MnO_4) is formed when
(1) Chlorine gas is passed into aq. $KMnO_4$ solution
(2) Potassium permanganate reacts with conc. H_2SO_4
(3) Manganese dioxide is fused with KOH in air
(4) $KMnO_4$ reacts with Cl_2 in acidic medium

83. The most stable oxidation states of Eu, Ce and Pm are respectively
(1) +2, +4, +3
(2) +4, +3, +2
(3) +4, +2, +3
(4) +3, +3, +3

84. H-bonding is absent in
(1) H_2O
(2) NH_3
(3) HCl
(4) CH_3OH

85. The correct order of hybridization of the central atom in the following species, SF_4, NH_4^+, $XeOF_4$, IF_7 respectively is
(1) sp^3d, sp^3, sp^3d^2, sp^3d^3
(2) sp^3d, dsp^2, sp^3d^2, sp^3d^3
(3) sp^3d^2, sp^3, sp^3d, sp^3d^3
(4) sp^3d, sp^3, sp^3d^2, sp^3d^2

SECTION-B

86. Which of the following is coloured due to *d-d* transition?
(1) $KMnO_4$
(2) Na_2CrO_4
(3) Cu_2O
(4) $[Ti(H_2O)_6]^{3+}$

87. Consider the following statements :
I. Enthalpy of atomisation of vanadium is maximum among 3d-series elements
II. Nickel is denser than iron
III. Co^{+3} ion can act as good oxidising agent for Cr^{+2} ion, reducing itself to Co^{+2} ion and oxidising Cr^{+2} to Cr^{+3}
The correct statement(s) is(are)
(1) Only (I) & (III)
(2) Only (III)
(3) Only (II)
(4) (I), (II) and (III)

88. $P_4 + NaOH + H_2O \rightarrow$ Products
Products are
(1) $H_3PO_4 + PH_3$
(2) $NaH_2PO_2 + PH_3$
(3) $Na_2HPO_2 + PH_3$
(4) $Na_3PO_4 + PH_3$

89. H_3PO_3 on heating gives
(1) H_3PO_4
(2) PH_3
(3) H_3PO_2
(4) Both (A) and (B)

90. In crown sulphur (S_8), number of 'S-S' bond(s) is/are
(1) 6
(2) 7
(3) 8
(4) 9

91. Compare the stability of the following species : H_2, H_2^+ and H_2^-
(1) $H_2^- < H_2^+ < H_2$
(2) $H_2 < H_2^+ < H_2^-$
(3) $H_2^+ < H_2^- < H_2$
(4) $H_2^+ < H_2 < H_2^-$

92. The bond order in Li_2 molecule is
(1) 1
(2) 1.5
(3) 2
(4) 2.5

93. The species which does not exist is
(1) H_2
(2) He_2
(3) C_2
(4) O_2

94. Given below are two statements one is labelled as Assertion (A) other is labelled as Reason (R)
Assertion (A) : B—F bond in BF_3 is polar.
Reason (R) : Dipole moment of BF_3 is zero.
In the light of above statements choose the correct answer
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

95. Given below are two statements one is labelled as Assertion (A) other is labelled as Reason (R)
Assertion (A) : $\overset{\oplus}{N}H_4$ is tetrahedral in shape.
Reason (R) : Bond angle H – N – H in $\overset{\oplus}{N}H_4$ is 109° 28'.
In the light of above statements choose the correct answer
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

96. Given below are two statements one is labelled as Assertion (A) other is labelled as Reason (R)
Assertion (A) : Sc^{3+} ion is colourless.
Reason (R) : Sc^{3+} ion has zero electron in d-subshell.
In the light of above statements choose the correct answer
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

97. Given below are two statements one is labelled as Assertion (A) other is labelled as Reason (R)
Assertion (A) : Zn shows +2 oxidation state in all of its compounds.
Reason (R) : Zn has completely filled d-subshell.
In the light of above statements choose the correct answer
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

98. XeF_4 on hydrolysis gives
(1) Xe
(2) XeO_3
(3) HF
(4) All of these

99. Select the incorrect statement regarding XeO_3.
(1) It is colourless compound
(2) Explosive in nature
(3) Physical state is solid
(4) Formed by the partial hydrolysis of XeF_6

100. Which of the following inter-halogen compounds does not exist?
(1) IF_7
(2) ClF_3
(3) ICl
(4) $BrCl_7$

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BOTANY

SECTION-A

101. Select the option in which all the taxa belongs to same taxonomic category?
(1) Primata, Insecta, Diptera
(2) Muscidae, Sapindales, Poaceae
(3) Mammalia, Dicotyledonae, Insecta
(4) Chordata, Hominidae, Poales

102. Keys are
(1) Used for identification of plants only
(2) Based on a pair of contrasting characters
(3) Generally theoretical in nature
(4) Same for all taxonomic categories

103. Select the incorrect option for binomial nomenclature
(1) Two components are generic name and specific epithet
(2) Specific epithet generally starts with a small letter
(3) Biological names are taken from latin language
(4) The name of the discoverer is printed in italics

104. Choose the incorrect match.
(1) Indian Botanical Garden – Situated in Lucknow
(2) Monographs – Contains information on any one taxon
(3) Museum – Contains preserved animals and plants
(4) Herbarium – Repository of plant specimens for future use

105. In which among the following taxonomic categories, organisms share less number of common characters than that of order?
(1) Family
(2) Species
(3) Class
(4) Genus

106. In mature pollen grain, generative cell differs from vegetative cell, as the former
(1) Has abundant food reserve
(2) Has a large irregularly shaped nucleus
(3) Has dense cytoplasm and it is spindle shaped
(4) Is bigger in size than latter

107. Match the column I with column II and select the correct option.

	Column I		Column II
a.	Pleiotropy	(i)	Human skin colour
b.	Polygenic inheritance	(ii)	Phenylketonuria
c.	Incomplete dominance	(iii)	Blood group 'AB'
d.	Co-dominance	(iv)	Gene for starch grain size in pea

(1) a(ii), b(i), c(iv), d(iii)
(2) a(iv), b(iii), c(ii), d(i)
(3) a(iii), b(iv), c(ii), d(i)
(4) a(ii), b(iii), c(iv), d(i)

108. Aristotle classified the animals on the basis of
(1) Natural affinities among them
(2) Phylogenetic relationships
(3) Absence or presence of red blood
(4) Morphological characters

109. Which of the following is not true for deuteromycetes?
(1) Asexually reproduce by conidia
(2) Mycelium is septate and unbranched
(3) Some members are saprophytic or parasite while a large number of them are decomposers of litter
(4) Commonly known as imperfect fungi

110. All of the following are the advantages of seeds to angiosperms, **except**
(1) Providing a rich food source for other organisms
(2) Better adaptive strategies for dispersal to new habitat
(3) Protection to young embryo
(4) Generation of new genetic combination

111. Cleistogamy refers to the condition in which
(1) Pollinators are not required
(2) Flowers are absent
(3) Flowers always remain open
(4) Flowers are gamopetalous

112. Amoeboid protozoans differ from all the other protozoan because they
(1) Use cilia for locomotion
(2) Have two types of nuclei
(3) Use flagella for movement
(4) Use pseudopodia for capturing prey

113. The basidiospores
(1) Are produced on basidium
(2) Are asexual spores
(3) Are diploid in nature
(4) Are endogenously produced spores

114. Choose the correct sequence of events with respect to fungal sexual cycle.
(1) Plasmogamy → Meiosis → Karyogamy
(2) Meiosis → Karyogamy → Plasmogamy
(3) Karyogamy → Plasmogamy → Meiosis
(4) Plasmogamy → Karyogamy → Meiosis

115. After entering the embryo sac, the pollen tube releases two male gametes into
(1) The cytoplasm of the central cell
(2) The antipodal cells
(3) The cytoplasm of the egg cell
(4) The cytoplasm of the synergids

116. In angiosperms, microsporogenesis is referred to
(1) Formation of pollen mother cell from microspore
(2) Formation of male gametes in the pollen grain
(3) Germination of pollen grain on the stigma
(4) Formation of microspores by meiotic division in pollen mother cell

117. For artificial hybridisation, emasculation is done if
(1) Female parent bears bisexual flowers
(2) Parent plants are dioecious
(3) Male parent bears unisexual flowers
(4) Flowers of female parent have all the sterile stamens

118. Read the following statements and select the **correct** ones for pollen grains.
(a) About 25-50 µm in diameter
(b) Surrounded by a single layered wall
(c) Intine has prominent germ pores due to absence of cellulose
(d) Are well preserved due to presence of sporopollenin
(1) a & d only
(2) a, b & d
(3) a, c & d
(4) c & d only

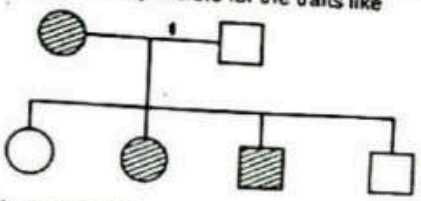
119. Hilum represents the junction between
(1) Funicle and ovary wall
(2) Ovule and funicle
(3) Chalaza and integument
(4) Funicle and micropyle

120. β-Thalassemia is controlled by a single gene HBB, which is present on chromosome
(1) 11
(2) 9
(3) 16
(4) 12

121. An exception to Mendelian principle, where both alleles express themselves equally in heterozygous condition is
(1) Incomplete dominance
(2) Codominance
(3) Pleiotropism
(4) Polygenic inheritance

122. Which of the given is **not** true for Down's syndrome?
(1) Occurs due to failure of segregation of chromatids of an autosome
(2) Affected individual have only one copy of chromosome 21
(3) Affected individual have short stature
(4) Its inheritance cannot be shown via pedigree chart

123. Study the given pedigree chart. This pedigree is not possible for the traits like



(1) Haemophilia
(2) Sickle cell anaemia
(3) Cystic fibrosis
(4) Phenylketonuria

124. Which among the following is not the basis of classification of kingdom fungi into various classes.
(1) Mode of spore formation
(2) Mode of nutrition
(3) Fruiting bodies
(4) Morphology of mycelium

125. The recessive allele
(1) May produce non-functional enzyme
(2) Is unmodified allele
(3) Produces normal enzyme
(4) Expresses itself only in heterozygous condition

126. Sickle cell anemia is due to replacement of A by B occurs at the sixth position of beta globin chain of haemoglobin molecule. Select the option that **correctly** fills A and B.
(1) **A**-Valine; **B**-Glutamic acid
(2) **A**-Phenylalanine; **B**-Tyrosine
(3) **A**-Tyrosine; **B**-Phenylalanine
(4) **A**-Glutamic acid; **B**-Valine

127. Causal agent of bovine spongiform encephalopathy
(1) Contains RNA as genetic material
(2) Is devoid of proteins
(3) Is abnormally folded protein
(4) Contains dsDNA

128. A woman has AB blood group. She marries to a man with blood group B whose mother had blood group O. Calculate the probability of their first child to be with blood group AB.
(1) $\frac{1}{16}$
(2) $\frac{1}{8}$
(3) $\frac{1}{2}$
(4) $\frac{1}{4}$

129. The term linkage was given by
(1) W. Sutton
(2) T. Boveri
(3) G. Mendel
(4) T.H. Morgan

130. Read the following statements.
(a) Code for a pair of contrasting traits
(b) They are slightly different forms of the same gene.
(c) True breeding line for a trait will have either both alleles dominant or both alleles recessive
(d) Both the alleles in a zygote are always contributed by one parent
Which ones are true for alleles?
(1) Only (a) & (d)
(2) Only (a), (b) & (c)
(3) Only (c) & (d)
(4) Only (b) & (d)

131. Select the option for those statements which are true for Punnett square.
(a) Developed by a British geneticist.
(b) Used to calculate the probability of all possible genotypes of offsprings.
(c) Useful for monohybrid crosses only.
(d) It is a mathematical formula which helps to calculate the types of gametes.
(1) (a), (b) and (c)
(2) (a) and (d)
(3) (b) and (d)
(4) Only (a) and (b)

132. Read the following statements and mark the **correct** option.
Statement A: Bacteria as a group show the most extensive metabolic diversity.
Statement B: Archaebacteria differ from other bacteria in having a different nuclear membrane structure and is responsible for their survival in extreme condition.
(1) Only statement A is correct
(2) Only statement B is correct
(3) Both the statements are correct
(4) Both the statements are incorrect

133. All of the following are common features between the members of Ascomycetes and Deuteromycetes, **except**
(1) Formation of conidia
(2) Can be decomposers
(3) Production of sexual spores
(4) Septate and branched mycelium

134. Function of endosperm is
(1) To provide nutrition to the developing embryo
(2) To give rise to the root apex
(3) To protect the egg cell
(4) To form the seed coat

135. Egg apparatus of embryo sac consists of
(1) Two egg cells and one synergid
(2) One egg cell and two synergids
(3) One egg cell and two antipodal cells
(4) One egg cell and one synergid

SECTION-B

136. In a cross performed by T.H Morgan considering eye colour and wings size in *Drosophila* what is the probability of males having miniature wings in F_2 generation?
(1) 62.8%
(2) 37.2%
(3) 1.3%
(4) 50%

137. Who united the knowledge of chromosomal segregation with Mendelian principles and called it the Chromosomal Theory of Inheritance?
(1) T.H. Morgan
(2) Sutton
(3) Mendel
(4) Correns

138. Identify the organisms on the basis of given information and select the **correct** option.
(i) They are the chief produces in the oceans.
(ii) Their walls are embedded with silica.
(1) Diatoms
(2) Mycoplasma
(3) Euglenoids
(4) Methanogens

139. Choose the **incorrect** option for sickle cell anemia.

(1) Individual with heterozygous genotype will be unaffected but will be the carrier of the disease

(2) Caused due to substitution at the sixth codon of beta globin gene from GUG to GAG

(3) This disease can be transmitted to offspring if both the parents are heterozygous for the gene

(4) The mutant haemoglobin molecule undergoes polymerization under low oxygen tension

140. Klinefelter's syndrome inflicted individuals

(1) Are sterile

(2) Show trisomy of chromosome 21

(3) Have karyotype of 45 + XO

(4) Are short statured with partially open mouth

141. Which among the following properties is exclusively present in the human beings?

(1) Self-consciousness

(2) Reproduction

(3) Cellular organisation

(4) Metabolism

142. All of the following are in the order Polymoniales, **except**

(1) Potato

(2) Datura

(3) Wheat

(4) *Petunia*

143. Choose the **incorrect** match.

(1) Syncarpous – Pistil are fused together

(2) Rice – Loose pollen viability within 30 minute of their release

(3) Antipodal cells – Situated at micropylar end

(4) Nucellus – Have abundant reserve food material

144. Which among the given plant features promotes autogamy but not geitonogamy or xenogamy?

(1) Monoecious condition

(2) Cleistogamy

(3) Dioecious condition

(4) Bisexual flowers

145. Match the following column I with column II and choose the **correct** option.

	Column I		Column II
a.	Scutellum	(i)	An undifferentiated sheath which encloses root cap
b.	Coleoptile	(ii)	A portion of embryonal axis above the level of cotyledon
c.	Coleorhiza	(iii)	Situated towards one side of the embryonal axis
d.	Epicotyl	(iv)	A hollow foliar structure which encloses shoot apex and leaf primordia

(1) a(iii), b(i), c(iv), d(ii)

(2) a(iii), b(i), c(ii), d(iv)

(3) a(ii), b(iv), c(i), d(iii)

(4) a(iii), b(iv), c(i), d(ii)

146. Read the following Assertion (A) and Reason (R) and then choose the **correct** option.

Assertion (A): A papaya plant prevents both autogamy and geitonogamy.

Reason (R): A papaya plant has either male flowers or female flowers but not both.

(1) Both A and R are true and R is correct explanation of A

(2) Both A and R are true and R is not a correct explanation of A

(3) Only A is true but R is false

(4) Both A and R are false

147. Choose the **correct** statement regarding the primary endosperm nucleus.

(1) It develops into embryo

(2) Its formation involves fusion of three gametes

(3) Its formation involves the fusion of three haploid nuclei

(4) It is a product of double fertilisation in gymnosperm

148. Identify the following statements as True (T) or False (F) and select the correct option.
A. Asexual reproduction in fungi is by spores, like oospores ascospores and basidiospores.
B. In *Albugo*, the mycelium is aseptate and coenocytic.
C. Aplanospores are non-motile asexual spores in phycomycetes.

	A	B	C
(1)	F	T	F
(2)	T	F	F
(3)	F	T	T
(4)	T	F	T

(1) (1)
(2) (2)
(3) (3)
(4) (4)

149. Male honeybee
(1) Is developed parthenogenetically
(2) Has same chromosome number as that of female bee
(3) Has father but not grandfather
(4) Is diploid

150. According to R.H. Whittaker, which character is absent in kingdom Animalia but present in kingdom Plantae?
(1) Eukaryotic cell types
(2) Cell wall
(3) Nuclear membrane
(4) Tissue body organisation

ZOOLOGY

SECTION-A

151. Which among the following is a connotation of 'theory of special creation' ?
(1) The Earth is about 4000 years old.
(2) The Universe is almost 20 billion years old.
(3) The atmosphere of primitive Earth was reducing in nature.
(4) The formation of life was preceded by chemical evolution

152. Select the **incorrect** statement.
(1) Key concepts of Darwinian theory of evolution are branching descent and natural selection
(2) Use and disuse of organs, inheritance of acquired characters are key concepts of Lamarckism
(3) Evolution for Darwin was gradual while for deVries, mutation caused speciation and hence called it saltation
(4) *Ramapithecus* was more ape-like while *Dryopithecus* was more man-like

153. Which of the following did **not** exist on primitive Earth in free form just before the origin of life?

(1) CO_2

(2) NH_3

(3) O_2

(4) CH_4

154. **Assertion (A)** : Before industrialization, there were more white winged moths on trees than melanised moths.
Reason (R) : Dark color in moths evolved due to a mutation induced by chemicals present in industrial smoke.
In the light of above statements, select the **correct** option

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion

(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

155. At high temperature, enzymes become __A__ while at low temperature enzymes remain in __B__ state. Select the **correct** option for A and B respectively.

(1) Inoperative, active

(2) Inactivated, active

(3) Denatured, inactive

(4) Inactivated, hyperactive

156. A common feature of peptide and glycosidic bond is that they

(1) Connect carbohydrate moieties

(2) Are formed by dehydration

(3) Are present in chitin

(4) Connect amino acids

157. How many amino acids and peptide bonds respectively are present in a pentapeptide?

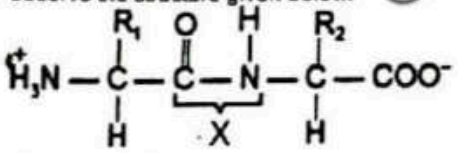
(1) 5, 5

(2) 5, 4

(3) 5, 3

(4) 4, 5

158. Observe the structure given below.



X represents a/an

(1) Ionic linkage

(2) Peptide bond

(3) Glycosidic bond

(4) Ester bond

159. Read the following statements w.r.t. co-factors and choose the **correct** option.
Statement A: Catalytic activity is lost when the co-factor is removed from the enzyme.
Statement B: Co-factors play a crucial role in the catalytic activity of the enzyme.

(1) Both statements A and B are correct

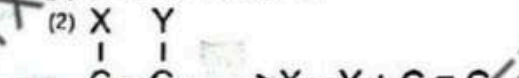
(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

160. An enzyme has enzyme commission number 2.1.1.6.
This enzyme belongs to the class of enzymes catalyzing which of the following reactions?

(1) $S - G + S' \rightarrow S + S' - G$

(2) 

(3) $CO_2 + H_2O \rightleftharpoons H_2CO_3$

(4) S reduced + S' oxidised → S oxidised + S' reduced

161. How many of the factors given in the box below will affect the activity of enzymes?

Temperature, pH, Substrate concentration, Enzyme concentration, Activation energy

(1) Five

(2) Four

(3) Three

(4) Two

162. Choose the **incorrect** match

(1)	Catalase	–	Haem is a prosthetic group and is a part of active site
(2)	Carboxypeptidase	–	Zinc is a cofactor which forms coordination bonds
(3)	Nicotinamide adenine dinucleotide	–	Contains vitamin niacin
(4)	Peroxidase	–	Contains hydrogen peroxide as prosthetic group

(1) (1)
(2) (2)
(3) (3)
(4) (4)

163. Below given are the features of a structure 'X' involved in the regulation of respiration.
(a) Neural signals from this centre can reduce the duration of inspiration.
(b) It can moderate the functions of respiratory rhythm centre.
(c) It is present in the pons region of the brain.
Select the option that correctly identifies 'X'.
(1) Pneumotaxic centre
(2) Respiratory rhythm centre
(3) Chemosensitive area
(4) Appetite centre

164. The partial pressure of carbon dioxide at the tissue level is
(1) More than that of pO_2 in systemic veins
(2) Equals to the pO_2 in pulmonary veins
(3) Less than that of pCO_2 in the atmosphere
(4) More than that of pO_2 at alveoli level

165. Match column I with column II and select the **correct** option.

	Column I		Column II
a.	Total lung capacity	(i)	ERV + TV + IRV
b.	Vital capacity	(ii)	ERV + RV
c.	Functional residual capacity	(iii)	RV + ERV + TV + IRV
d.	Expiratory capacity	(iv)	TV + ERV

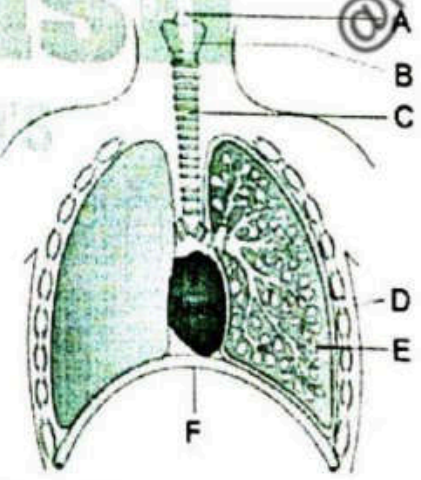
(1) a(iii), b(ii), c(iv), d(i)
(2) a(iv), b(ii), c(i), d(iii)
(3) a(iii), b(i), c(ii), d(iv)
(4) a(i), b(iii), c(iv), d(ii)

166. Which of the following phenomenon occurring during inspiration, is responsible in causing an increase in the volume of the thoracic chamber in the dorsoventral axis?
(1) Contraction of external inter-costal muscles
(2) Contraction of diaphragm
(3) Contraction of internal inter-costal muscles
(4) Relaxation of diaphragm

167. Conditions that favours the dissociation of oxygen from the oxyhaemoglobin are
(1) High pCO_2, High pO_2
(2) High H^+ concentration, High pCO_2
(3) High temperature, Low H^+ concentration
(4) High pO_2, High pH

168. Select the option which includes the conducting parts of human respiratory system.
(1) External nostrils, *terminal bronchioles*
(2) Secondary bronchi, alveolar duct
(3) Tertiary bronchi, alveoli
(4) Terminal bronchioles, alveoli

169. The figure shows diagrammatic representation of human lungs along with other respiratory structures. Select the option that gives the **correct** match of the labelling with its function.



(1) E → It is covered by a double layered pleura with pleural fluid in between
(2) F → It is in relaxed state during inspiration
(3) C → Pharynx opens through the larynx into this region
(4) A → It is a bony box which helps in sound production

170. All of the following exchange O_2 with CO_2 by simple diffusion over their entire body surface, **except**
(1) *Spongilla*
(2) *Hydra*
(3) *Planaria*
(4) *Periplaneta*

171. Organism 'X' belongs to the second largest phylum of kingdom Animalia whereas organism 'Y' is a vertebrate, characterized by the creeping or crawling mode of locomotion. The habitat of 'X' is aquatic. Identify 'X' and 'Y' and select the preferred mode of respiration of them respectively.
(1) Branchial respiration and cutaneous respiration
(2) Buccopharyngeal respiration and pulmonary respiration
(3) Branchial respiration and pulmonary respiration
(4) Cutaneous respiration and buccopharyngeal respiration

172. Which among the following structures is the part of conducting zone of respiratory system in humans but is not supported by the incomplete cartilaginous rings?
(1) Initial bronchioles
(2) Tertiary bronchi
(3) Alveolar ducts
(4) Terminal bronchioles

173. **Assertion (A)** : With the increase in substrate concentration, the velocity of the enzymatic reaction rises at first, the reaction ultimately reaches a maximum velocity (V_{max}) which is not exceeded by any further rise in concentration of the substrate.
Reason: (R) The enzyme molecules are fewer than the substrate molecules and after saturation of the molecules, there are no free enzyme molecules to bind with the additional substrate molecules.
In the light of above statements, choose the correct option.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) Both (A) and (R) are false
(4) (A) is true and (R) is false

174. The numerical value of which of the following pulmonary capacities is least in a healthy human at rest?
(1) Expiratory capacity
(2) Functional residual capacity
(3) Inspiratory capacity
(4) Vital capacity

175. The total amount of CO_2 transported under normal conditions in humans through blood plasma is
(1) 7 %
(2) 22 %
(3) 77 %
(4) 70 %

176. Consider the reaction given below:

$$CO_2 + H_2O \overset{1}{\rightleftarrows} H_2CO_3 \overset{2}{\rightleftarrows} HCO_3^- + H^+$$

Select the **correct** option.
(1) The reaction proceeds towards the right direction when the levels of the carbondioxide are low
(2) The enzyme catalysing the reaction 1 belongs to the VIth class of enzymes
(3) The above reaction occurs mainly in the blood plasma.
(4) The reaction proceeds in the reverse direction at the site where exchange of gases occurs in between the blood and atmospheric air

177. In a hypothetical population of 1000 people, tests of blood type genes show that 160 have genotype AA, 480 have the genotype AB and 360 have the genotype BB. Calculate the frequency of allele B and select the **correct** option.
(1) 0.40
(2) 0.60
(3) 0.10
(4) 0.02

178. All of the following are examples of evolution due to involvement of *Homo sapiens*, **except**
(1) Decrease in the number of white winged moths post-industrialization
(2) Development of drug resistant microbes
(3) Variations in the beaks of finches of Galapagos islands
(4) Herbicide resistance varieties of crops

179. Darwin's finches are an example of
(a) Natural selection
(b) Divergent evolution
(c) Adaptive radiation
(d) Analogy
Select the **correct** option.
(1) (a), (b) and (c)
(2) (b), (c) and (d)
(3) (a), (c) and (d)
(4) (a), (b) and (d)

180. Which among the following pairs show convergent evolution ?

(1)	Banded anteater	–	Koala
(2)	Marsupial mouse	–	Bandicoot
(3)	Flying squirrel	–	Flying phalanger
(4)	Spotted cuscus	–	Tiger cat

(1) (1)
(2) (2)
(3) (3)
(4) (4)

181. Most enzymes get damaged at high temperatures above 40°C. However, enzymes isolated from organisms who normally live under extremely high temperatures can retain their catalytic power even at high temperatures upto
(1) 80-90°C
(2) 140-150°C
(3) 180-190°C
(4) 200-210°C

182. A colony of bacteria containing two different varieties *i.e.*, 'A' and 'B' is surviving in an island since million years. During the course of evolution, the temperature of the island rises due to global warming. Later, it was observed by the scientists that the population of 'B' outgrowths the population of 'A'.
What conclusion among the following can be made from this observation?
(1) Population 'A' has complete dominance over population 'B'
(2) Population 'A' is reproductively more fit than that of 'B'
(3) Fitness of 'B' is better than 'A' under the new conditions
(4) Population of 'A' can survive better in hostile environment

183. Which of the following is responsible for the movement of O_2 from the alveoli into the blood in the pulmonary capillaries?
(1) Active transport
(2) Facilitated diffusion
(3) Simple diffusion
(4) Osmosis

184. A French naturalist, even before Darwin, explained his thoughts about the mechanism of evolution by giving the example of giraffes who in an attempt to forage leaves on tall trees had to adapt by elongation of their neck. He proposed the theory of
(1) Natural selection
(2) Inheritance of acquired characters
(3) Panspermia
(4) Spontaneous generation

185. Select the **incorrect** option w.r.t. Malthusian theory that influenced Darwin.
(1) Population are nearly stable in size, except seasonal fluctuations.
(2) Natural resources are limited in nature.
(3) Every population has an inherent capacity to increase its number logarithmically if every individual of that population reproduced maximally
(4) Nature keeps a check of population by posing several challenges like diseases, food shortages, etc.

SECTION-B

186. What will be the total lung capacity of a person with tidal volume of 450 mL, vital capacity of 4000 mL, and a residual volume of 1000mL?
(1) 4500 mL
(2) 5000 mL
(3) 5500 mL
(4) 6000 mL

187. Read the following statements and select the **correct** option.
Statement A: The rate of diffusion of gases is inversely related to the thickness of diffusion membrane.
Statement B: Negative pressure breathing is characteristic of mammals.
(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

188. The volume of air remaining in the lungs even after a forcible expiration cannot be measured by the spirometer.
Which among the following set of capacities include this volume?
(1) TLC and VC
(2) FRC and IC
(3) TLC and FRC
(4) VC and EC

189. Every 2 L of oxygenated blood can deliver about 'X' mL of oxygen to the tissues under normal conditions in humans. Choose the **correct** value for 'X'.
(1) 80
(2) 100
(3) 12
(4) 120

190. After holding the breath for a while, we feel the urge to breath due to the
(1) Increase in concentration of oxygen
(2) Increase in the pH of blood
(3) Increase in the carbon dioxide concentration
(4) Decrease in H^+ concentration

191. The partial pressure of oxygen in all of the following blood vessels is greater than that of vena cava, **except**
(1) *Thoracic aorta*
(2) Pulmonary artery
(3) Pulmonary vein
(4) Hepatic artery

192. In the section of a forest, there are more medium sized plants than tall plants and dwarf plants. Over a period of time, it was observed that the number of dwarf and tall plants is twice as that of medium sized plants.
It is an example of
(1) Disruptive selection
(2) Directional selection
(3) Stabilising selection
(4) Balanced selection

193. Rhynia-like plants gave rise to 'X'. 'X' is a genus of extinct land plants of Devonian period.
All of the following evolved from 'X', **except**
(1) Sphenopsids
(2) Ginkgos
(3) Gnetales
(4) Zosterophyllum

194. Select the **incorrect** statement.
(1) The jawless fish probably evolved around 350 mya.
(2) Lobefins that belong to the Osteichthyes are thought to be evolved into the first amphibians
(3) *Crocodilus* is more closely related to Aves than that of *Chelone*.
(4) Amphibians lay thick-shelled eggs which do not dry up in sun unlike those of reptiles

195. Consider the following features:
(a) Discovered in Java
(b) Probably were carnivores
(c) Existed about 1.5 mya
The above mentioned features are related to
(1) *Homo habilis*
(2) Neanderthal man
(3) *Homo erectus*
(4) *Ramapithecus*

196. Comprehend the features given below w.r.t. dinosaurs.
(a) Herbivore in nature
(b) Plated back and spikes at the end of tail
The above mentioned features are associated with
(1) *Tyrannosaurus*
(2) *Stegosaurus*
(3) *Archaeopteryx*
(4) *Pteranodon*

197. In adult human haemoglobin, four polypeptides interact with each other to attain optimal activity by forming which of the following structures of proteins?
(1) Primary
(2) Secondary
(3) Tertiary
(4) Quaternary

198. Enzymes catalysing removal of groups from substrates by mechanism other than hydrolysis leaving double bonds are categorised in which class as per IUB.
(1) Class II
(2) Class III
(3) Class IV
(4) Class VI

199. Read the following statements and select the **correct** option.
Statement A: All enzymes are proteinaceous in nature.
Statement B: Enzymes majorly exhibit quaternary structures which lack secondary and primary structures in any conditions.
(1) Both statements A and B are incorrect
(2) Only statement A is correct
(3) Only statement B is correct
(4) Both statements A and B are correct

200. During an enzymatic reaction, a transition state structure of the substrate is formed which is
(1) Stable and transient
(2) Unstable and transient
(3) Stable and permanent
(4) Unstable and permanent

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